

Faris Hittiny

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EDUCATION

Texas A&M University
Bachelor of Computer Engineering, GPA: 3.65

College Station, TX
Expected May 2027

EXPERIENCE

AI/ML Undergraduate Research Assistant - Design and Development of AI/Machine Learning Models for Disease Management in Animal Systems May 2025 - Present
College Station, TX

- Processed and labeled >10,000 images from 4+ months of continuous video, creating scalable datasets for training/validation.
- Developed and benchmarked 10+ deep learning models (YOLO, Mask R-CNN, Detectron2), achieving ~95% accuracy in object identification.
- Automated pipelines that reduced manual preprocessing time by >60%, improving dataset scalability.
- Produced reproducible results through documented workflows, version control (Git), and HPC scheduling (Slurm).
- Collaborated in an 11-member research team, preparing models for real-time animal health monitoring systems.

IEEE Student Member - Texas A&M University August 2024 - Present

- Completed workshops in embedded AI (Jetson/Raspberry Pi), Linux tools, and PCB design.
- Engaged with 400+ peers and 10+ industry leaders (NASA, Micron) at the TAMU IEEE Summit, discussing AI hardware and IT infrastructure.

PROJECTS

AI-Driven Demand Mapping – Tidal Hackathon (Top 10)

- Built a Python/AWS ML pipeline analyzing 50,000+ sales records to recommend warehouse locations for Amazon sellers.
- Visualized geospatial + sales-trend data, improving delivery efficiency by an estimated 20%.
- Ranked Top 10 out of 200+ teams, presenting results in a 4-member Agile team.

RISC-V Accelerator on Zephyr

github.com/FarisHittiny

- Designed a 6-register accelerator in Verilog with AXI-Lite interface, integrated into LiteX SoC.
- Built and validated a Zephyr RTOS driver through 100+ Renode/Verilator simulations.
- Benchmarked checksum workloads, achieving a 3× speedup over software-only execution.
- Demonstrated hardware acceleration potential for AI/ML pipelines and documented results for reproducibility.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, YOLOv8, Detectron2, OpenCV, scikit-learn, NumPy, Pandas

Programming & Systems: Python, C++, C, Bash, MATLAB, Git, Linux, Docker, Slurm, SSH, CI/CD

Cloud & IT Tools: AWS, Jupyter, VS Code, LaTeX, WSL

Embedded & Acceleration: Verilog, FPGA prototyping, LiteX, Renode, Verilator, Zephyr RTOS

Data & Visualization: Pandas, NumPy, Matplotlib